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Unit – 3 $\Rightarrow$ Probability Distribution

Introduction

→ Suppose an AI weather model is predicting tomorrow's weather.

→ If I ask:

What is the probability that tomorrow will be rainy?

→ The model answers:

$$P(\text{Rain}) = 0.7$$

→ This tells us the probability of one event only.

→ But what if I want the complete picture of tomorrow's weather?

Weather Tomorrow	Probability
Sunny	0.1
Cloudy	0.2
Rainy	0.6
Stormy	0.1

→ Now I know:

- All possible weather conditions that may occur tomorrow.
- The probability of each weather condition.

→ This complete description is called a **Probability Distribution**.

Probability Distribution

→ A **Probability Distribution** is a table, graph, or mathematical function that shows all possible values (or outcomes) of a random variable and the probability associated with each value.

- A probability tells us the chance of one outcome.
- A probability distribution tells us the chances of all possible outcomes.

Unit – 3 Probability Distribution

- In this chapter, we will study some important probability distributions, their parameters, and their applications. We will learn when and where these distributions can be used to model real-life random phenomena and support decision-making.
- Imagine two questions:
 - How many students will be absent today?
 - How long will you wait for the next bus?
- Both involve uncertainty — but they are not the same kind of uncertainty.
 - In the first case, the answer can only be 0, 1, 2, 3, ... — we can **count** the outcomes.
 - In the second case, the answer could be 5.2 minutes, 5.25 minutes, or 5.251 minutes — we **measure** the outcome.
- This difference leads to two powerful ideas in probability:

(1) Discrete Probability Distributions (PMF)

- When outcomes are countable, we use Discrete Probability Distributions, described by a Probability Mass Function (PMF).
- PMF assigns a probability to each countable value of a random variable.
- To model such situations, we study:
 - (1) **Binomial Distribution** – counts successes in repeated trials
 - (2) **Poisson Distribution** – counts occurrences of events in a fixed interval

(2) Continuous Probability Distributions (PDF)

- When outcomes are measurable and can take any value within an interval, we use Continuous Probability Distributions, described by a Probability Density Function (PDF).
- PDF describes how probability is distributed over a range of values.
- To model such situations, we study:
 - (3) **Exponential Distribution** – models waiting time until the next event
 - (4) **Normal Distribution** – models measurements that cluster around an average

Method 1 $\rightsquigarrow$ Binomial Distribution

→ Before studying Binomial Distribution, what kind of experiment does it model?

- The answer is Bernoulli Trial.

Bernoulli Trials

→ A Bernoulli's trial is a random experiment that satisfies all of the following conditions:

(1) There are only two possible outcomes

(1) Success (S)

(2) Failure (F)

- “**Success**” does not mean something good. It simply means the outcome we are interested in.
- Example: Consider a coin toss.
 - There are only two outcomes: Head (H) and Tail (T)
 - Suppose we are interested in getting a Head.
 - Then:

Success (S) = Head and Failure (F) = Tail

(2) The probability of success remains the same in each trial.

i.e., Probability of success is constant.

- For a fair coin: $P(\text{Head}) = 0.5$
- If Head is defined as Success, then $P(S) = 0.5$ on every toss.
- The probability does not change from one trial to another.
- Whether it is the 1st toss, 5th toss, or 100th toss, the probability of getting a Head remains 0.5.

(3) Each trial is independent.

- Suppose the first toss results in a Head.

What is the probability of getting a Head on the second toss?

- Still, $P(\text{Head}) = 0.5$
- The first toss does not affect the second toss.
- Similarly, the second toss does not affect the third toss.

Unit – 3 Probability Distribution

→ Therefore, a coin toss is a Bernoulli Trial because it has exactly two outcomes, constant probability of success, and independent repetitions.

→ **Remark**

- Let $P(\text{Success}) = p$; $0 \leq p \leq 1$ then $P(\text{Failure}) = 1 - p = q$
($\because p + q = 1 \rightarrow$ Mutually Exclusive and Exhaustive)

Bernoulli Random Variable

→ A Bernoulli Random Variable is a random variable associated with a Bernoulli trial that takes only two values: 1 (Success) and 0 (Failure).

OR

→ Let random variable X represent the outcome of a Bernoulli trial:

$$X = \begin{cases} 1 & \text{; if success occurs} \\ 0 & \text{; if failure occurs} \end{cases}$$

→ Then

$$P(X = 1) = p \text{ and } P(X = 0) = 1 - p = q$$

→ This random variable is called a Bernoulli Random Variable.

→ Note that, a Bernoulli Random Variable describes the outcome of a single Bernoulli trial.

→ But in many real-life situations, we do not perform the experiment just once.

- Tossing a coin 10 times
- Testing 100 products
- Sending 50 emails
- Predicting whether 100 students pass or fail

Binomial Distribution

→ A Bernoulli Random Variable describes the outcome of a single trial.

→ When a Bernoulli trial is repeated n times and we count the number of successes, we obtain a **Binomial Random Variable X**.

→ The probability distribution of X is called the **Binomial Distribution**.

Unit – 3 Probability Distribution

→ Definition

- The Binomial Distribution is the probability distribution of a random variable that represents the number of successes in n independent Bernoulli trials, where each trial has the same probability of success p .

Binomial Random Variable

→ Let

X = Number of successes in n Bernoulli trials

→ Then X is called a Binomial Random Variable.

→ If each trial has probability of success p , then

$$X \sim B(n, p)$$

[read as: X follows a Binomial Distribution with parameters n and p]

where

- n = Number of trials
- p = Probability of success in each trial
- $q = 1 - p$ = Probability of failure

→ Examples:

- (1) A box contains 100 bolts. If each bolt has a probability p of being defective, what is the probability that exactly 5 bolts are defective?
- (2) A coin is tossed 10 times. What is the probability of obtaining at least 6 heads?
- (3) In the next 20 births at a hospital, what is the probability that at most 12 babies are female?

→ To solve such probability questions, we use the Binomial Probability Mass Function (PMF).

→ The **probability mass function** of a Binomial Random Variable is

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x} ; x = 0, 1, 2, \dots, n$$

Unit – 3 Probability Distribution

Conditions for a Binomial Experiment

→ An experiment is Binomial if all these conditions are satisfied:

- (1) **Fixed** number of trials (n is fixed in advance) --- **F**
- (2) Each trial has only **two** outcomes (success / failure) --- **T**
- (3) Trials are **independent** --- **I**
- (4) Probability of success (p) remains **constant**/does not change from trial to trial --- **C**

→ If any one condition fails, it is not Binomial.

Properties of Binomial Distribution

(1) The mean of binomial distribution is defined as

$$\mu = E(X) = np$$

$$E(X) = \sum x \cdot P(X = x) = np$$

(2) The **variance** of binomial distribution is defined as $V(X) = npq$.

(3) The **standard deviation** of binomial distribution is defined as $\sigma = \sqrt{npq}$.

Example of Method-1: Binomial Distribution

H	1	What conditions must be satisfied for a random variable to follow a Binomial Distribution? Answer: Refer the Concepts
C	2	For the binomial distribution with $n = 20$, $p = 0.35$. Find mean, variance and standard deviation. Answer: Mean = 7, Variance = 4.55, Standard Deviation = 2.1331
C	3	Obtain the binomial distribution function for which mean is 10 and variance is 5. Answer: $P(X = x) = \binom{20}{x} (0.5)^x (0.5)^{20-x}$; $x = 0, 1, 2, \dots, 20$.

Unit – 3 Probability Distribution

H	4	Find the binomial distribution for $n = 4$ and $p = 0.3$. Answer: <table><tr><td>X</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>P(X)</td><td>0.2401</td><td>0.4116</td><td>0.2646</td><td>0.0756</td><td>0.0081</td></tr></table>	X	0	1	2	3	4	P(X)	0.2401	0.4116	0.2646	0.0756	0.0081
X	0	1	2	3	4									
P(X)	0.2401	0.4116	0.2646	0.0756	0.0081									
H	5	Let $X \sim B(6, p)$. Given that $4 \cdot P(X = 4) = 5 \cdot P(X = 3)$, determine the value of the probability of success p. Answer: $p = 0.625$												
H	6	If the probability that a bolt is defective is 0.1, find the mean and standard deviation of the number of defective bolts in a total of 400 bolts. Answer: Mean = 40, Standard Deviation = 6												
H	7	A machine produces items with a probability $p = 0.02$ of being defective. If 100 items are inspected, find the expected number of defective items. Answer: 2												
T	8	A random variable X follows a Binomial Distribution. The mean of the distribution is 12 and the variance is 4.8. Find: (1) The values of n and p. (2) The standard deviation. (3) $P(X = 10)$. (4) $P(X \geq 10)$. Answer: (1) $n = 20, p = 0.6$, (2) $\sigma \approx 2.191$, (3) 0.1171, (4) 0.8725												
C	9	12% of the tablets produced by a tablet machine are defective. What is the probability that out of a random sample of 20 tablets produced by the machine, 5 are defective? Answer: 0.0567												

Unit – 3 Probability Distribution

H	10	In a city, the probability that a randomly selected car driver does not carry a driving license is 0.25. If a traffic inspector randomly checks 3 drivers, find the probability that exactly one driver is not carrying a driving license. Answer: 0.4219
T	11	Find the probability that in a family of 4 children there will be at least 1 boy. Assume that the probability of a male birth is 0.5. Answer: 0.9375
H	12	20% of the bulbs produced by a factory are defective. If 4 bulbs are selected at random, find the probability that at most 2 bulbs are defective. Answer: 0.9728
T	13	A disease occurs in 5% of the population. If 20 people are selected at random, find the probability that exactly one person has the disease. Answer: 0.3774
H	14	A university warehouse receives a large shipment of printers in which 40% are laser printers and 60% are inkjet printers. If 6 printers are selected at random, assuming each selection is independent, find the probability that exactly 3 are laser printers. Answer: 0.2765
H	15	The failure rate in a certain examination is 40%. If 6 candidates are selected at random, find the probability that at least 4 candidates pass the examination. Answer: 0.5443
C	16	A pair of fair dice is thrown 3 times. Find the probability of obtaining a sum of 7 at least once. Answer: $\frac{91}{216}$

Unit – 3 Probability Distribution

H	17	Each water sample has a 10% probability of containing a particular organic pollutant. Assuming the samples are independent, find the probability that at least 4 out of the next 18 samples contain the pollutant. Answer: 0.0979
H	18	Ten fair coins are tossed simultaneously. Find the probability of obtaining at least 7 heads. Answer: 0.1719
C	19	The probability that India wins a Test match against Australia is $\frac{1}{3}$ Assuming the outcomes of the matches are independent, if India and Australia play 3 Test matches, find the probability that: (a) India loses all three Test matches. (b) India wins at least one Test match. Answer: (a) 0.2963, (b) 0.7037
T	20	Assume that, on average, one out of every fifteen telephone numbers called between 1 p.m. and 2 p.m. on weekdays is busy. If 6 telephone numbers are selected at random, find the probability that: (a) Not more than 3 telephone numbers are busy. (b) At least 3 telephone numbers are busy. Answer: (a) 0.9997, (b) 0.0051
H	21	The probability that a randomly selected student in a university is a postgraduate is 0.6. If 8 students are selected independently, find the probability that: (a) None of the students is a postgraduate. (b) Exactly two students are postgraduates. (c) At least two students are postgraduates. Answer: 0.0007, 0.0413, 0.9915

Unit – 3 Probability Distribution

H	22	The probability that an infection is cured by a particular antibiotic drug within 5 days is 0.75. Assuming the treatment outcomes of different patients are independent, 4 patients are treated with the drug. Find the probability that: (a) No patient is cured. (b) Exactly two patients are cured. (c) At least two patients are cured. Answer: 0.0039, 0.2109, 0.9492
C	23	By assuming equal probabilities for boys and girls. Out of 800 families with 4 children each, how many would you expect to have (1) 2 boys and 2 girls? (2) at least 1 boy? (3) at most 2 girls? (4) no girl? Answer: (1) 300, (2) 750, (3) 550, (4) 50
H	24	Assume that the probability of the birth of a boy and a girl is equal. Out of 2000 families, each having 4 children, determine the expected number of families having: (a) At least 1 boy. (b) Exactly 2 boys. (c) 1 or 2 girls. (d) No girls. Answer: (a) 1875, (b) 750, (c) 1250, (d) 125
C	25	A fair die is tossed 5 times. Find the probability that the face 3 appears: (a) Exactly twice. (b) At most once. (c) At least two times. Answer: (a) $\frac{625}{3888}$, (b) $\frac{3125}{3888}$, (c) $\frac{763}{3888}$

Unit – 3 Probability Distribution

H	26	A fair die is thrown 6 times. Let a success be obtaining an odd number. Find the probability of: (a) Exactly 5 successes. (b) At least 5 successes. (c) At most 5 successes. Answer: (a) $\frac{3}{32}$, (b) $\frac{7}{64}$, (c) $\frac{63}{64}$
T	27	A multiple-choice test consists of 8 questions, each having 3 answer choices, of which only one is correct. A student selects an answer by rolling a fair die: he chooses the first answer if the outcome is 1 or 2, the second answer if it is 3 or 4, and the third answer if it is 5 or 6. If a distinction requires at least 75% correct answers and there is no negative marking, find the probability that the student secures a distinction. Answer: 0.0197
T	28	A large e-commerce company uses an AI-powered recommendation engine. Historical data shows that the probability of the recommendation engine recommending the correct product to a user is 0.9. Assuming recommendations for different users are independent, a quality assurance team randomly selects 20 user sessions. Find the probability that: (1) Exactly 18 recommendations are correct. (2) At least 18 recommendations are correct. (3) The recommendation engine makes at most 2 incorrect recommendations. (4) Interpret the results in terms of customer satisfaction and system reliability. Answer: (1) 0.2852, (2) 0.6770, (3) 0.6770

T	29	A cricket franchise uses an AI-based performance analysis system to predict whether a batsman will score 30 or more runs in a match. Historical data shows that the system predicts correctly with probability 0.8. Assuming the predictions for different batsmen are independent, the AI evaluates 15 batsmen before the IPL auction. Find: (1) The probability that the AI predicts the performance of exactly 12 batsmen correctly. (2) The probability that it predicts the performance of at least 13 batsmen correctly. (3) The expected number of correct predictions. (4) The standard deviation of the number of correct predictions. (5) Interpret the results in terms of the reliability of the AI system for player selection. Answer: (1) 0.2501, (2) 0.3980, (3) 12, (4) 1.5492
T	30	A streaming platform distributes incoming traffic across 10 identical and independent cloud server instances. The platform operates normally as long as at least 8 servers remain online (i.e., at most 2 servers crash). If 3 or more servers crash, the platform experiences a major outage. Historical data shows that each server crashes independently with probability 0.05 during peak load. Find: (1) The expected number of server crashes. (2) The probability that the platform handles the peak traffic without an outage. Answer: (1) 0.5, (2) 98.84%

Method 2 $\rightsquigarrow$ Poisson Distribution

Introduction

- A company's server receives, on average, 3 cyber-attacks per hour.
 - What is the probability that exactly 5 attacks occur in the next hour?
- Problems involving the number of occurrences of an event in a fixed interval of time or space are modeled using the Poisson Distribution.
- Poisson distribution answers one simple but powerful question:
If events occur randomly at an average rate λ , how many times will the event occur in a fixed interval?
- Interval can be:
 - Time (per minute, per hour, per day)
 - Space (per square meter, per kilometer)
 - Volume / area / length
- Examples
 - Number of phone calls received in 10 minutes.
 - Number of accidents at a crossing in one day.
 - Number of login requests to a server in 1 minute.
 - Number of cyber-attacks detected in 1 hour.
 - Number of defects in 1 meter of cloth.
 - Number of customers entering a shop in 1 hour.
- Here,
 - time, length, area → these are continuous
 - number of occurrences/events → this is countable
- Poisson is not modeling the interval.
- Poisson is modeling the count of events inside the interval.
 - So, the random variable is discrete.

Unit – 3 Probability Distribution

Poisson Distribution

→ A Poisson Distribution is a discrete probability distribution that describes the probability of a given number of events occurring in a fixed interval of time, space, area, or volume, when the events occur independently and at a constant average rate.

→ Example

- Number of emails received per hour.
- Number of customers arriving at a store per minute.
- Number of network failures per day.
- Number of printing mistakes on a page.
- Number of accidents at a traffic signal per month.

Poisson Random Variable

→ Let

X = Number of events occurring in a fixed interval

Then X is called a Poisson Random Variable.

→ If the average number of occurrences is λ , then

$$X \sim \text{Poisson}(\lambda) \text{ or } X \sim P(\lambda)$$

where

λ = average number of events in the fixed interval

→ Its probability mass function is given by

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}; x = 0, 1, 2, 3, \dots$$

Poisson Parameter λ

→ If events occur at an average rate r per unit, then the Poisson parameter for an interval of length L is $\lambda = r \times L$.

- i.e. If Rate = r events per unit
- Interval = L unit

$$\therefore \lambda = \frac{r}{\text{unit}} \times L \text{ unit} = r \times L \rightarrow \text{unit less}$$

→ Before computing λ , express the interval length in the same unit as the denominator of the given rate.

Unit – 3 Probability Distribution

Why is Poisson called the Distribution of Rare Events?

- Imagine a company has a cybersecurity system that receives 1,00,000 login attempts in a day.
- For one particular login attempt, suppose the probability that it is a malicious attack is very small:

$$P(\text{attack}) = 0.00001$$

- So, for one login attempt, an attack is very unlikely.
- This is what we mean by a “rare event.”
- The system receives 1,00,000 login attempts, so there are 1,00,000 opportunities for this rare event to occur.
- The cybersecurity team may not be interested only in whether one particular login is an attack. Instead, they may want to know:

How many suspicious attacks will occur during the next hour?

- Now we are counting the occurrences of a rare event over a fixed interval.
- This is where the **Poisson distribution** becomes useful.
- An event is called **rare** because it is unlikely to happen at a particular moment.
- Poisson distribution tells us how many times that rare event occurs during a given time interval.

Poisson is called the distribution of rare events because it tells us how many times an unlikely event occurs in a fixed interval of time or space.

Relationship with the Binomial Distribution

- Poisson distribution is the **limiting case** of Binomial distribution.
- When
 - Number of trials n is very large [$n \rightarrow \infty$]
 - Probability of success p is very small [$p \rightarrow 0$]
 - But $np = \lambda$ remains finite

$$\therefore \text{Binomial}(n, p) \approx \text{Poisson}(\lambda)$$

- This explains why Poisson is called “**distribution of rare events**”.

Unit – 3 Probability Distribution

Properties of Poisson Distribution

→ The mean and variance of the Poisson distribution with parameter λ are defined as

$$\text{mean } \mu = E(X) = \lambda = np \quad \& \quad \text{variance } V(X) = \sigma^2 = \lambda.$$

Example of Method-2: Poisson Distribution

C	1	If the random variable X follows a Poisson distribution and $P(X = 3) = P(X = 4)$, then find $P(X = 0)$. Answer: $P(X = 0) = e^{-4}$																
H	2	If the random variable X follows a Poisson distribution and $P(X = 1) = P(X = 2)$, then find mean and standard deviation of this distribution. Also, find $P(X = 3)$. Answer: Mean = 2 , SD = $\sqrt{2}$, P(X = 3) = 0.1804																
H	3	A random variable X follows a Poisson Distribution. If $\frac{3}{2} \cdot P(X = 1) = P(X = 3),$ find $P(X \geq 1)$. Answer: ≈ 0.9502																
T	4	Why is the Poisson distribution often called the "distribution of rare events"? Explain with suitable examples. Answer: Refer the Concept																
C	5	The number of page requests received by a web server per second is modeled as a Poisson random variable X . Its probability distribution is given below. <table><tr><td>Number of requests/sec. x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>Probability $P(X = x)$</td><td>0.368</td><td>0.368</td><td>0.184</td><td>0.061</td><td>0.015</td><td>0.003</td><td>0.001</td></tr></table> Find the mean and variance of X. Answer: $E(X) = 1$, V(X) = 1	Number of requests/sec. x	0	1	2	3	4	5	6	Probability $P(X = x)$	0.368	0.368	0.184	0.061	0.015	0.003	0.001
Number of requests/sec. x	0	1	2	3	4	5	6											
Probability $P(X = x)$	0.368	0.368	0.184	0.061	0.015	0.003	0.001											

Unit – 3 Probability Distribution

H	6	The probability that a wafer contains a large contaminating particle is 0.01. Assume that the wafers are independent. If 15 wafers are analyzed, find the probability that no wafer contains a large contaminating particle. Answer: 0.8607
H	5	A book of 585 pages contains 43 typographical errors, which are randomly distributed throughout the book. Find the probability that 10 randomly selected pages are free from typographical errors. Answer: 0.4795
T	6	A book contains 100 misprints randomly distributed throughout its 100 pages. Find the probability that a randomly selected page contains at least two misprints. Answer: 0.2642
C	7	A large number of samples, each containing 20 parts, are taken from the output of a machine. The mean number of defective parts per sample is 2. Out of 1000 such samples, how many would be expected to contain exactly 2 defective parts? Answer: 271
H	8	A lot contains 5000 electric bulbs, of which 100 are defective. Assuming the defective bulbs are randomly distributed, find the probability that a box of 100 bulbs contains at most 3 defective bulbs. Answer: ≈ 0.8571
C	9	The number of potholes on a highway follows a Poisson distribution with an average of 2 potholes per mile. Find the probability that no more than 4 potholes occur in a 5-mile section of the highway. [$e^{-10} \approx 0.0000454$] Answer: ≈ 0.0293

Unit – 3 Probability Distribution

T	10	A company has 250 workers. The probability that a worker is absent on any given day is 0.02. Assuming the absences of workers are independent, find the probability that exactly 7 workers are absent on a particular day. [$e^{-5} \approx 0.006738$] Answer: ≈ 0.1044
H	11	The number of monthly breakdowns of a computer follows a Poisson distribution with mean 1.8. Find the probability that the computer functions for a month: (a) Without any breakdown. (b) With at least one breakdown. Answer: (a) 0.1653, (b) 0.8347
H	12	In a bolt manufacturing company, the probability that a bolt is defective is $\frac{1}{500}$. The bolts are packed in packets of 20 bolts each. Using the Poisson distribution, find the approximate number of packets containing: (a) No defective bolts. (b) Exactly two defective bolts. Assume there are 10,000 packets in the consignment. Answer: (a) ≈ 9608, (b) ≈ 8
H	13	The average number of accidents on a national highway is 1.8 per day. Assuming that the number of accidents follows a Poisson distribution, find the probability that the number of accidents in a day is: (a) At least 1. (b) At most 1. [$e^{-1.8} \approx 0.1653$] Answer: (a) 0.8347, (b) 0.4628

Unit – 3 Probability Distribution

T	14	The number of flaws in a fiber optic cable follows a Poisson process with an average of 0.6 flaw per 100 feet. (a) Find the probability of exactly 2 flaws in a 200-foot cable. (b) Find the probability of exactly 1 flaw in the first 100 feet and exactly 1 flaw in the second 100 feet. [$e^{-0.6} \approx 0.5488$, $e^{-1.2} \approx 0.3010$] Answer: (a) 0.2167, (b) 0.1084
H	15	A publisher ensures that the probability of any page containing at least one typographical error is 0.005. Assume that pages are independent. A novel contains 400 pages. Using the Poisson approximation, find the probability that the novel contains: (a) Exactly one page with errors. (b) At most three pages with errors. Answer: (a) 0.2707, (b) 0.8571
H	16	The probability that an individual suffers a bad reaction from a certain injection is 0.001. Assuming the reactions are independent, find the probability that out of 2000 individuals: (a) More than 2 individuals suffer a bad reaction. (b) Exactly 3 individuals suffer a bad reaction. Answer: (a) 0.3233, (b) 0.1804
C	17	The probability that a person catches the coronavirus is 0.001. Assuming that infections are independent, find the probability that out of 3000 persons: (a) Exactly 3 persons catch the virus. (b) More than 2 persons catch the virus. [$e^{-3} \approx 0.0498$] Answer: (a) 0.2240, (b) 0.5768

Unit – 3 Probability Distribution

T	18	In a factory manufacturing razor blade, the probability that a blade is defective is 0.002. The blades are packed in packets of 10. Using the Poisson distribution, calculate the approximate number of packets containing: (a) No defective blade. (b) Exactly one defective blade. (c) Exactly two defective blades. The consignment contains 10,000 packets. [$e^{-0.02} \approx 0.9802$] Answer: (a) 9802, (b) 196, (c) 2
H	19	A car hire firm has two cars, which are hired out on a daily basis. The number of daily demands for a car follows a Poisson distribution with mean 1.5. Find the proportion of days on which: (a) Neither car is used. (b) Some demand is refused. Answer: (a) 0.2231, (b) 0.1912
C	20	A machine produces items, 1% of which are defective. Using the Poisson approximation, find the probability that a sample of 100 items contains: (a) No defective items (all good items). (b) Exactly one defective item. (c) At least three defective items. Answer: (a) 0.3679, (b) 0.3679, (c) 0.0803
H	21	A cybersecurity team receives ransomware alerts according to a Poisson distribution with an average rate of 1.5 alerts per hour. The team can investigate at most 3 alerts per hour. If more than 3 alerts are received in an hour, the team cannot investigate all of them and is considered overwhelmed. Find the probability that the team becomes overwhelmed during the next hour. Answer: 6.56%

Unit – 3 Probability Distribution

T	22	A tech influencer receives negative comments according to a Poisson distribution with an average rate of 3 comments per hour after posting a controversial AI opinion. Find the probability that: (1) Exactly 5 negative comments arrive in the next hour. (2) No negative comments arrive in the next hour. (3) The influencer deletes the post if more than 6 negative comments arrive in an hour. What is the probability that the post gets deleted? [$e^{-3} \approx 0.0498$] Answer: (1) 0.1008, (2) 0.0498, (3) 0.0335
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T	23	A spacecraft experiences micro-meteorite impacts at an average rate of 0.2 impacts per hour. The spacecraft is scheduled to travel through a debris field for 10 hours. The number of impacts during this period can be modeled using a Poisson distribution. The current shielding system can safely withstand at most 2 impacts. (1) What is the probability that the spacecraft experiences exactly 2 impacts during the mission segment? (2) What is the probability that the shielding system fails? (3) Based on the failure probability, would you approve the mission launch? Justify your answer. (4) Engineers propose upgrading the shielding system so that it can safely withstand up to 4 impacts instead of 2. - What will be the new probability of shielding failure? (5) As the mission safety officer, would you recommend: - Launching with the original shield or upgrading the shield? Support your recommendation using probability calculations. Answer: (1) 27.07%, (2) 32.33%, (3) The mission should not be approved with the current shield. (4) 5.27% (5) Upgrade Upgrade the shielding system before launch. as it reduces the failure probability from 32.33% to 5.27%
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Unit – 3 Probability Distribution

T	24	During an IPL final, the total number of sixes hit by both teams follows a Poisson distribution with an average of 12 sixes per match. Before the match, the franchise owner announces: <i>"If 18 or more sixes are hit during today's final, every one of the 50,000 spectators will receive an official team jersey free of cost."</i> Each jersey costs ₹500. Find: (1) The probability that exactly 18 sixes are hit. (2) The probability that the franchise distributes free jerseys to all spectators. (3) The expected promotional cost of the campaign. (4) If the marketing budget is ₹1 crore, comment on whether the owner's announcement is financially sensible. (5) If the owner wants the probability of distributing jerseys to be less than 5%, determine the minimum qualifying number of sixes. Answer: (1) 2.55%, (2) 6.30%, (3) around ₹15.74 lakhs (4) Mathematically: Yes. Financially (Risk Management): No. (5) 19
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Method 3 $\rightsquigarrow$ Exponential Distribution

Introduction

→ Consider a web server that receives user requests randomly throughout the day.

- Suppose the server receives, on average, 50 requests per minute.
- Using the **Poisson Distribution**, we can answer questions such as:

How many requests will arrive in the next minute?

- This is a **counting problem**, so we use the Poisson Distribution.

→ But now consider a different question:

- Suppose the server is currently idle.
 - A system administrator asks:

How long will it take until the next request arrives?

→ Notice that we are no longer counting requests.

→ Instead, we are measuring the **waiting time** until the next request arrives.

→ This leads to the **Exponential Distribution**.

Exponential Random Variable

→ Let

X = Waiting time until the next occurrence of an event

Then X is called an **Exponential Random Variable**.

→ Examples:

- Time until the next user request arrives at a web server.
- Time until the next email is received.
- Time until the next customer enters a store.
- Time until the next network failure occurs.

→ Since waiting time can take any **non-negative** real value, X is a continuous random variable.

$$X \geq 0$$

Unit – 3 Probability Distribution

Exponential Distribution

→ An **Exponential Distribution** is a continuous probability distribution that describes the waiting time until the next occurrence of an event.

→ Example

- Suppose a cybersecurity system detects, on average, **3 attacks per hour**.
- Poisson Distribution answers:

How many attacks will occur in the next hour?

- Exponential Distribution answers:

How much time will pass before the next attack occurs?

→ If the average rate of occurrence is r events per unit time, then the random variable X is said to follow an Exponential Distribution and is denoted by

$$X \sim \text{Exp}(r)$$

Interpretation of r

→ Suppose a server receives $r = 5$ requests per minute.

→ Then Exponential Distribution can answer questions such as:

- What is the probability that the next request arrives within 10 seconds?
- What is the probability that we wait more than 30 seconds for the next request?

→ Poisson counts events, whereas Exponential measures the waiting time between events.

Probability Density Function (PDF) of Exponential Distribution

→ Let

X = Waiting time until the next occurrence of an event

and let r be the average rate of occurrence of events.

→ Then the Probability Density Function (PDF) of X is

$$f(x) = \begin{cases} r \cdot e^{-rx} ; x \geq 0 \\ 0 ; x < 0 \end{cases}$$

where, **mean** = $\frac{1}{r}$ **unit** and **variance** = $\frac{1}{r^2}$ (**unit**)².

Unit – 3 Probability Distribution

→ In exponential distribution we can find the probability as given below:

$$(1) P(X \leq x) = 1 - e^{-rx}$$

$$(2) P(X \geq x) = e^{-rx}$$

$$(3) P(a \leq X \leq b) = e^{-ar} - e^{-br}$$

The Memoryless Property

→ **What does "memoryless" mean?**

- Imagine a computer program that generates a notification at random times.
- You have been waiting 5 minutes and still haven't received one.
- A student might think: *"Now the notification must be coming soon!"*
 - Not necessarily!
- For an exponential distribution, the time already waited does not affect the future waiting time.
- This is called the memoryless property.
- The process has no memory of how long you have already waited.

→ **Mathematical Statement**

If $X \sim \text{Exp}(r)$, then for $s, t > 0$

$$P(X > s + t \mid X > s) = P(X > t).$$

→ **Interpretation:**

- If you have already waited s minutes, the probability of waiting another t minutes is the same as if you had just started waiting.

→ **Example:**

- Suppose buses arrive according to an exponential waiting-time model.
- You are told:

The probability that you wait more than 10 minutes is 0.30.

- Now imagine you have **already waited 10 minutes** and the bus still hasn't arrived.
- What is the probability that you will wait **at least 15 minutes**?

Unit – 3 Probability Distribution

- By the memoryless property:

$$P(X > 15 + 10 | X > 10) = P(X > 10)$$

- Therefore,

$$P(X > 25 | X > 10) = 0.3$$

- So even after waiting 10 minutes, your probability of waiting another 15 minutes is still **30%**.

→ The distribution does not care how long you have already waited.

→ That's why it is called memoryless.

→ **Derivation**

- The Exponential distribution is memoryless.

Proof:

- For an exponential random variable, $P(X > x) = e^{-rx}$
- Now,

$$\begin{aligned} P(X > s + t | X > s) &= \frac{P(X > s + t) \cap P(X > s)}{P(X > s)} \\ &= \frac{P(X > s + t)}{P(X > s)} \\ &= \frac{e^{-r(s+t)}}{e^{-rs}} = e^{-rt} \end{aligned}$$

- But $e^{-rt} = P(X > t)$
- Hence,

$$P(X > s + t | X > s) = P(X > t)$$

Example of Method-3: Exponential Distribution

C	1	The lifetime T of an alkaline battery is exponentially distributed with $r = 0.05$ per hour. What are mean and standard deviation of batteries lifetime? Answer: $E(T) = 20$ hours, $\sigma(T) = 20$ hours
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Unit – 3 Probability Distribution

C	2	A random variable has an exponential distribution with probability density $f(x) = \begin{cases} \frac{1}{5} e^{-\frac{x}{5}} & ; x > 0 \\ 0 & ; \text{otherwise} \end{cases}$ Find (a) $P(X > 5)$, (b) $P(3 \leq X \leq 6)$, (c) Mean, (d) Variance Answer: (a) 0.3679, (b) 0.2478, (c) 5, (d) 25
H	3	A random variable X has an exponential distribution with probability density $f(x) = \begin{cases} \frac{1}{3} e^{-\frac{x}{3}} & ; x > 0 \\ 0 & ; \text{otherwise} \end{cases}$ Find (a) $P(X > 2)$, (b) $P(2 < X < 5)$, (c) Mean and Variance Answer: (a) 0.5134, (b) 0.3245, (c) $E(X) = 3, V(X) = 9$
T	4	A random variable has an exponential distribution with probability density $f(x) = \begin{cases} ce^{-2x} & ; x > 0 \\ 0 & ; \text{otherwise} \end{cases}$ Find value of the constant c and $P(X > 5)$. Answer: c = 2, 0.0000454
H	5	The lifetime T (in hours) of an alkaline battery follows an Exponential Distribution with rate parameter $\theta = 0.02$ per hour. Find: (a) The mean lifetime. (b) The variance of the battery lifetime. Answer: $E(T) = 50$ hours, $V(T) = 2500$ hours²
H	6	The time required to repair a machine follows an Exponential Distribution with rate parameter $r = 0.5$ per hour. Find the probability that the repair time exceeds 3 hours. Answer: 0.2231

Unit – 3 Probability Distribution

T	7	The lifetime T (in hours) of a machine component follows an exponential distribution with rate parameter r. It is known that $P(T > 20) = 0.3679$. Find: (a) The value of r. (b) The mean lifetime. (c) The variance. (d) $P(10 < T < 30)$. (e) Given that the component has already survived 20 hours, find the probability that it survives for at least another 10 hours. Answer: (a) $r = 0.05$ per hour, (b) $E(T) = 20$ hours (c) $V(T) = 400$ hours², (d) 0.3834, (e) 0.6065
H	8	The time interval T (in minutes) between two successive barges passing a certain point on a busy waterway follows an Exponential Distribution with a mean of 8 minutes. Find the probability that the time interval between two successive barges is less than 5 minutes. Answer: 0.4647
C	9	The time between breakdowns of a particular machine follows an Exponential Distribution with a mean of 17 days. Find the probability that the machine breaks down within the next 15 days. Answer: 0.5863
C	10	The lifetime T (in hours) of an alkaline battery follows an Exponential Distribution with rate parameter $r = 0.05$ per hour. Find: (a) The probability that the battery lasts between 10 and 15 hours. (b) The probability that the battery lasts more than 20 hours. Answer: (a) 0.1341, (b) 0.3679

Unit – 3 Probability Distribution

H	11	The lifetime T (in years) of a computer part follows an Exponential Distribution with a mean lifetime of 10 years. Find: (a) The probability that the computer part lasts more than 7 years. (b) The probability that the computer part lasts between 9 and 11 years. Answer: (a) 0.4966, (b) 0.0737
H	12	The mileage X obtained by car owners from a certain type of radial tire follows an Exponential Distribution with a mean of 4000 km. Find the probability that a tire will last: (a) at least 2000 km (b) at most 3000 km. Answer: (a) 0.6065, (b) 0.5276
C	13	In a large corporate computer network, user log-ons to the system can be modeled as a Poisson process with a mean rate of 25 log-ons per hour. (a) What is the probability that there are no log-ons during a 6-minute interval? (b) What is the probability that the time until the next log-on is between 2 and 3 minutes? Answer: (a) 0.0821, (b) 0.1481
T	14	Accidents occur according to a Poisson process at an average rate of 4 accidents per week. (a) Calculate the probability of more than 5 accidents in a given week. (b) Find the probability that at least 2 weeks elapse between two successive accidents. Answer: (a) 0.2149, (b) 0.000335

Unit – 3 Probability Distribution

H	15	The arrival rate of cars at a gas station is 40 cars per hour. Assume that arrivals follow a Poisson process. (a) What is the probability of having no arrivals during a 5-minute interval? (b) What is the probability of having exactly 3 arrivals during a 5-minute interval? Answer: (a) 0.0357, (b) 0.2202
T	16	The length of time X required to complete a job is exponentially distributed with $E(X) = \mu = \frac{1}{\lambda} = 10 \text{ hours.}$ (a) Find the probability that the time required to complete a job exceeds 20 hours. (b) The cost of completing a job is given by $C = 4 + 2X + 2X^2$. Find the expected cost $E(C)$. Answer: (a) 0.1353, (b) 424
H	17	A random variable X has an exponential distribution with probability density function $f(x) = \begin{cases} 3e^{-3x} & ; x > 0 \\ 0 & ; x \leq 0 \end{cases}$ Find the probability that X is not less than 4. Answer: e^{-12}
C	18	The income tax X (in thousands of rupees) paid by a person follows an exponential distribution with probability density function $f(x) = \begin{cases} \frac{1}{3} e^{-\left(\frac{x}{3}\right)} & ; x > 0 \\ 0 & ; \text{otherwise} \end{cases}$ The income tax is levied at the rate of 15% on the portion of income exceeding ₹15,000. Find the probability that the person's annual income exceeds ₹17,000. Answer: 0.9048

Unit – 3 Probability Distribution

T	19	The lifetime X of an electronic component follows an Exponential Distribution with a mean lifetime of 4 years. The manufacturer provides a replacement warranty of 3 years. (a) What proportion of components will be replaced during the warranty period? (b) What is the probability that the lifetime of a randomly selected component lies within two standard deviations of the mean lifetime? Answer: (a) 0.5276, (b) 0.9502
C	20	Marvel Studios unexpectedly releases teaser trailers during a live fan event. The waiting time until the next surprise trailer follows an Exponential Distribution with a rate of 0.5 trailers per hour. Fans have already watched the event for 90 minutes without any trailer. (1) What is the probability that they must wait at least another 20 minutes for the next trailer? (2) A fan shouts, <i>"We've waited this long! The trailer must be coming next!"</i> Using the memoryless property of the Exponential Distribution, determine whether this statement is mathematically correct. Answer: (1) 84.65%, (2) The fan's statement is mathematically incorrect due to memoryless property.
H	21	A computer program generates notifications at random times. The waiting time X follows an exponential distribution. If you have already waited 5 minutes without receiving a notification, what is the probability that you will wait at least another 3 minutes, given that $P(X > 3) = 0.4$? Answer: 0.4.

T	22	During a football tournament, the waiting time T until the first VAR review follows an exponential distribution with a rate of 0.4 reviews per match-hour. (1) What is the probability that the first VAR review occurs within the first 20 minutes? (2) What is the probability that no VAR review occurs during the first half (45 minutes)? (3) If no VAR review has occurred during the first half, what is the probability that no VAR review occurs during the next 30 minutes? (4) Explain whether the absence of a VAR review in the first half changes the likelihood of a VAR review occurring later. Answer: (1) 0.1248, (2) 0.7408, (3) 0.8187 (4) No, it does not change the likelihood at all, as the exponential distribution is memoryless.
T	23	A cybersecurity company monitors a global network. Based on historical observations, successful zero-day attacks reaching their monitoring system occur randomly at an average rate of 2 attacks per week. The SOC (Security Operations Center) operates 24×7. (1) What is the probability that the first successful attack is detected within the first two days? (2) What is the probability that more than one week passes before the first successful attack? (3) If no successful attack has occurred during the first week, what is the probability that the team must wait more than another four days? (4) Compute the expected waiting time until the next successful attack. Answer: (1) 0.4353, (2) 0.1353, (3) 0.3189 (4) 0.5 week or 3.5 days

Unit – 3 Probability Distribution

T	24	A deep-space observatory monitors near-Earth asteroids. Based on decades of observations, potentially hazardous asteroid detections occur randomly at an average rate of 0.4 detections per month. A planetary defense team is on high alert during the next mission. (1) What is the probability that the first hazardous asteroid is detected within the next 20 days? (2) What is the probability that the team enjoys at least four months without any hazardous asteroid detection? (3) The mission has already completed five months without any detection. What is the probability that the team will remain detection-free for at least another two months? (4) A new engineer claims, <i>"Since we have survived five months without detecting any hazardous asteroid, a detection is now overdue."</i> Is this statement mathematically correct? Justify your answer. Answer: (1) 0.2341, (2) 0.2019, (3) 0.4493 (4) No, the engineer's statement is mathematically incorrect.
T	25	A smartphone manufacturer studies sudden battery failures after a software update. The waiting time until the next unexpected battery failure follows an Exponential Distribution with a rate of 0.02 failures per day. A user's phone has been working perfectly for the last 100 days. (1) What is the probability that the phone will continue to work for at least another 30 days without a battery failure? (2) The user says, <i>"My phone has survived 100 days, so it is bound to fail soon."</i> Using the memoryless property of the exponential distribution, determine whether this statement is correct. Answer: (1) 0.5488, (2) No. The statement is incorrect because of the memoryless property..

Unit – 3 Probability Distribution

T	26	In cryptocurrency markets, a whale is an individual or organization that owns a very large amount of cryptocurrency. A whale transaction is a very large buy or sell order that can significantly affect the market price. A cryptocurrency exchange has found that whale transactions occur randomly at an average rate of 1.2 transactions per hour. (1) What is the probability that the first whale transaction occurs within the next 20 minutes? (2) What is the probability that traders observe no whale transaction for more than 2 hours? (3) After monitoring the market for 90 minutes without any whale transaction, what is the probability that traders must wait at least another 30 minutes before the next whale transaction occurs? (4) An investor claims: <i>“Since there has been no whale transaction for the last 90 minutes, one is now overdue and is more likely to occur very soon.”</i> Using the Exponential Distribution, determine whether this claim is valid and explain your reasoning. Answer: (1) 0.3297, (2) 0.0907, (3) 0.5488 (4) The claim is incorrect. The exponential distribution is memoryless, so waiting 90 minutes does not make a whale transaction "overdue."
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Unit – 3 Probability Distribution

T	27	A university announces semester examination results at random times after the examinations are completed. Historical records suggest that the waiting time until the result announcement follows an Exponential Distribution with a rate of 0.5 announcements per week. Students have already waited three weeks without the results being announced. (1) What is the probability that the results are not announced during the next one week? (2) A student comments: <i>“We’ve already waited three weeks, so the result announcement is now overdue and is more likely to happen tomorrow.”</i> Using the memoryless property of the Exponential Distribution, determine whether this statement is mathematically correct. Answer: (1) 60.65%, (2) The student’s statement is mathematically incorrect due to memoryless property.
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Method 4 $\rightsquigarrow$ Normal Distribution

Introduction

→ We already know:

- **Binomial Distribution** → Counts success/failure outcomes.
- **Poisson Distribution** → Counts how many times an event occurs.
- **Exponential Distribution** → Measures waiting time until an event occurs.

Notice something interesting. All three distributions answer very specific questions.

- Binomial asks 'How many successes?'
- Poisson asks 'How many events?'
- Exponential asks 'How long must we wait?'

But does every real-life situation fit into one of these three categories?

→ In reality, we often observe values such as:

- Heights of students
- Marks in an examination
- Measurement errors
- Sensor readings
- IQ scores
- Blood pressure
- Temperature

→ These are neither counts nor waiting times.

→ **These values:**

- Are continuous in nature.
- Can take many possible values.
- Are influenced by several factors simultaneously.
- Do not arise from a single cause.

→ **Example:**

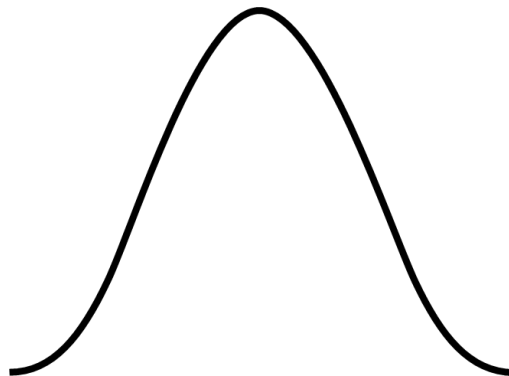
- A student's mark depends on:
 - Understanding of concepts

Unit – 3 Probability Distribution

- Memory
 - Preparation
 - Sleep
 - Health
 - Stress level
 - Question difficulty
 - Teaching quality
 - Luck
 - Transition
- No single factor determines the final mark.
 - Instead, many small factors work together to produce the final outcome.
 - As a result, observations naturally cluster around an average value.
 - This creates a characteristic shape known as the **Bell Curve**.

Bell Curve

- A mathematical model describing how values are distributed around an average when many small random factors act together.
- In simple words:
 - Most observations lie near the average.
 - Fewer observations lie far from the average.
 - Extreme observations are rare.
- This produces the familiar bell-shaped curve.



Unit – 3 Probability Distribution

But why does this bell shape appear in so many completely different situations?

→ The most important idea in statistics:

- ***When many independent random variables are added, their sum tends to follow a Normal Distribution — regardless of the original distribution.***

→ Because this pattern appears so frequently in nature, science, engineering, and data analysis, statisticians developed a mathematical model to describe it.

Normal Distribution

→ The Normal Distribution is a continuous probability distribution that describes how values are distributed around a central value when many small random factors act together.

→ It is characterized by:

- Mean (μ)
- Standard Deviation (σ)

and has the bell-shaped curve.

→ **Why is it called "Normal"?**

- Normal Distribution
 - introduced by mathematicians studying measurement errors.
 - found repeatedly in nature and experiments.
 - became so common that it was called the "Normal" distribution.

Definition

→ A continuous random variable X is said to follow a Normal Distribution with mean μ and variance σ^2 , written as $X \sim N(\mu, \sigma^2)$, if its probability density function is given by,

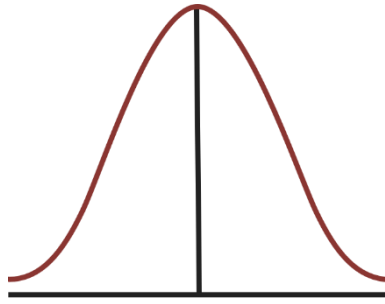
$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} \exp \left[-\frac{1}{2} \left(\frac{x - \mu}{\sigma} \right)^2 \right]$$

where, $-\infty < x < \infty, -\infty < \mu < \infty$ & $\sigma > 0$

→ Here, μ (mean) & σ^2 (variance) are known as parameters of the distribution.

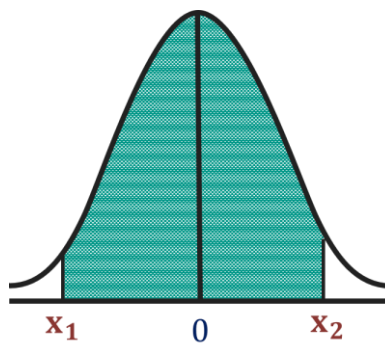
→ The curve representing the normal distribution is called the normal curve.

Unit – 3 Probability Distribution



→ Note that

- The probability that X lies between x_1 and x_2 is equal to the area under the normal curve between $x = x_1$ and $x = x_2$



→ However, a practical problem remains:

- A Normal random variable is written as

$$X \sim N(\mu, \sigma^2)$$

- Now imagine this:
 - Heights of students $\rightarrow X \sim N(165, 102)$
 - Exam scores $\rightarrow Y \sim N(60, 152)$

→ But the problem is different normal distributions have different values of μ and σ , so each has

- Different curves
- Different probability calculations
- Infinite varieties

→ If we compute probabilities separately for every Normal Distribution, we would need:

- Different calculations

Unit – 3 Probability Distribution

- Different tables
- Repeated integrations

which is impractical.

Can we somehow convert every Normal Distribution into a single common form?

→ Note that

- Although their centers and spreads differ, all Normal Distributions have a similar bell-shaped appearance.
- They only differ in location (μ) and scale (σ)

→ So, the natural question is:

- Can we shift and rescale any normal variable so that all normals look identical?
 - Yes. By subtracting the mean and dividing by the standard deviation, any Normal random variable can be transformed into a common variable called the Standard Normal Variable **Z**.

→ **What does "Standardizing" actually mean?**

- Standardization is a two-step transformation:

Step 1: Shift

$$X - \mu$$

- Subtract the mean.
- This moves the centre from μ to **0**.

Step 2: Scale

$$\frac{X - \mu}{\sigma}$$

- Divide by the standard deviation.
- This makes the standard deviation equal to 1.

→ So, we define:

$$Z = \frac{X - \mu}{\sigma}$$

→ This new variable is called the Standard Normal Variable.

Unit – 3 Probability Distribution

- So, every normal random variable is transformed into the same standard normal distribution after standardization.
- Real-life normal distributions differ in mean and spread.
 - Z-score standardization removes these differences by rescaling data, so all normal distributions map to the same standard normal form without changing their relative structure.

Standard Normal Variable

- A standard normal variable **Z** is a normal random variable with: **$Z \sim N(0, 1)$** .
- Standard Normal Variable is defined as

$$Z = \frac{X - \mu}{\sigma}$$

- That means:
- Mean **$E(Z) = 0$**
 - Standard deviation **$\sigma(Z) = 1$**

→ Interpretation of Z-score

$$Z = \frac{X - \mu}{\sigma}$$

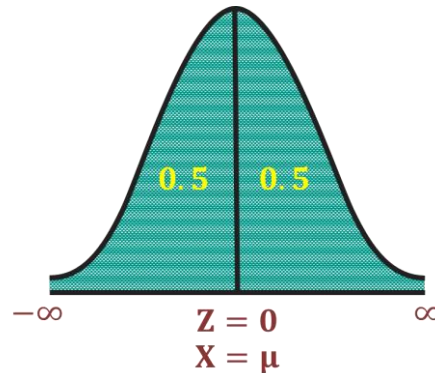
- tells us
- How many standard deviations an observation lies above or below the mean.
- Examples:
- $Z = 0 \rightarrow$ exactly at the mean
 - $Z = 1 \rightarrow$ one standard deviation above mean
 - $Z = -2 \rightarrow$ two standard deviations below mean
- PDF of Standard Normal Variable Z is

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} ; -\infty < z < \infty$$

Unit – 3 Probability Distribution

Methods for Finding the Probability(area) from Z - Table

- The normal curve represents the probability density of a normal distribution.
- The total area under the curve (from $-\infty$ to $+\infty$) is 1.



- The curve is symmetric about the mean μ (or $Z = 0$).
- The probability(area) to the right side of $X = \mu$ is 0.5 and is denoted by
 $P(X \geq \mu) = 0.5$
- The probability(area) to the left side of $X = \mu$ is also 0.5 and is denoted by
 $P(X \leq \mu) = 0.5$

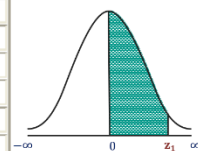
Important Cases

(1) **Area between $Z = 0$ to $Z = z_1$, which can be represented as $P(0 \leq Z \leq z_1)$.**

- Find z_1 by using first row and column of Z - table to get the answer of $P(0 \leq Z \leq z_1)$.
- Example:

- Suppose $z_1 = 1.43$. Use below steps to find value at $z = 1.43$.
 - Step 1: Split 1.43 as 1.4 + 0.03.
 - Step 2: Find 1.4 in first column and 0.03 in first row.

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441



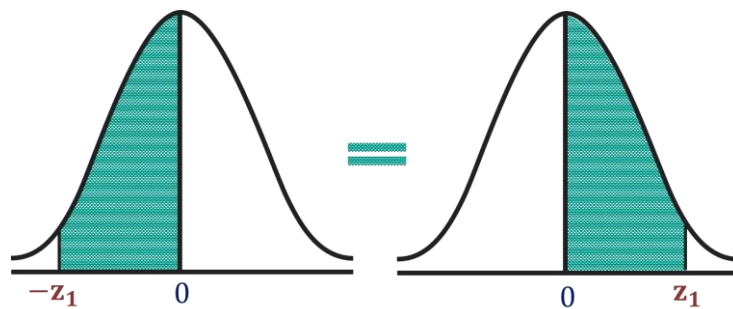
Unit – 3 Probability Distribution

- Step 3: Intersection of that row and column is answer of $P(0 \leq Z \leq 1.43)$.
- Here, $P(0 \leq Z \leq 1.43) = 0.4236$

→ **Note:**

- Area between $Z = -z_1$ to $Z = 0$, which can be represented as $P(-z_1 \leq Z \leq 0)$.
- So, $P(-z_1 \leq Z \leq 0) = P(0 \leq Z \leq z_1)$ as normal distribution is **symmetric** about x – axis.

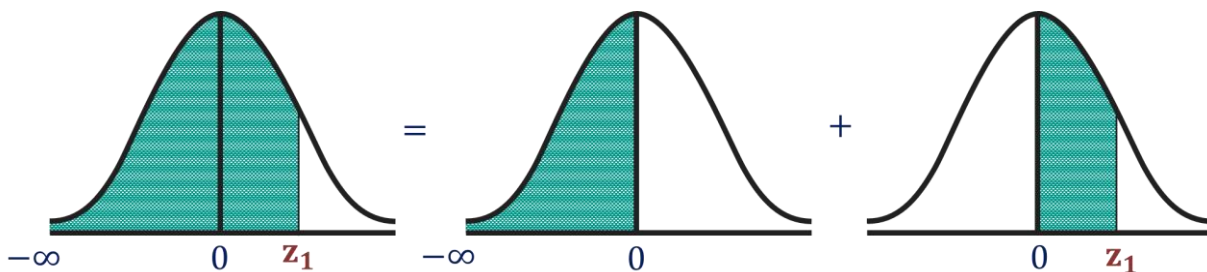
That is, $P(0 \leq Z \leq 1.43) = P(-1.43 \leq Z \leq 0) = 0.4236$



(2) Area between $Z = -\infty$ to $Z = z_1$, which can be represented as $P(Z \leq z_1)$.

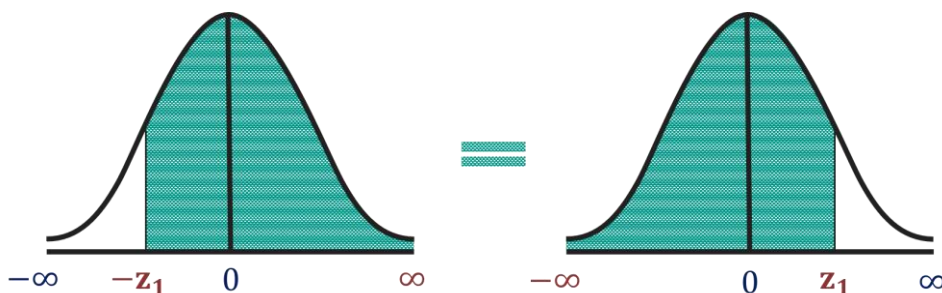
→ Now, $P(Z \leq z_1) = P(-\infty \leq Z \leq 0) + P(0 \leq Z \leq z_1)$

→ Since, $P(-\infty \leq Z \leq 0) = 0.5$ and $P(0 \leq Z \leq z_1)$ can be found by using case – I.



→ **Note:**

- Area between $Z = -z_1$ to $Z = \infty$ which can be represented as $P(Z \geq -z_1)$
 - Also, $P(Z \geq -z_1) = P(Z \leq z_1)$ as normal distribution is **symmetric** about x – axis

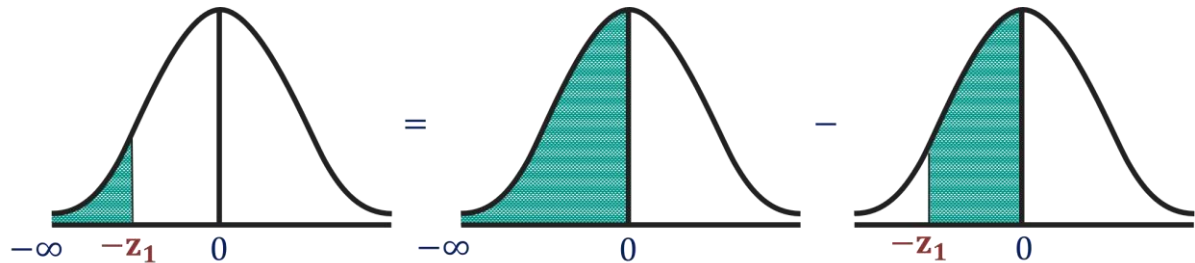


Unit – 3 Probability Distribution

(3) **Area between $Z = -\infty$ to $Z = -z_1$, which can be represented as $P(Z \leq -z_1)$**

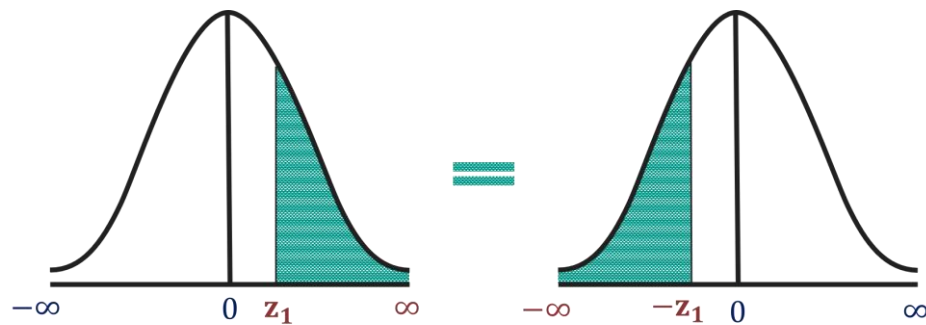
→ Now, $P(Z \leq -z_1) = P(-\infty \leq Z \leq 0) - P(-z_1 \leq Z \leq 0)$

→ Since, $P(-\infty \leq Z \leq 0) = 0.5$ and $P(-z_1 \leq Z \leq 0)$ can be found by using case – I.



→ **Note:**

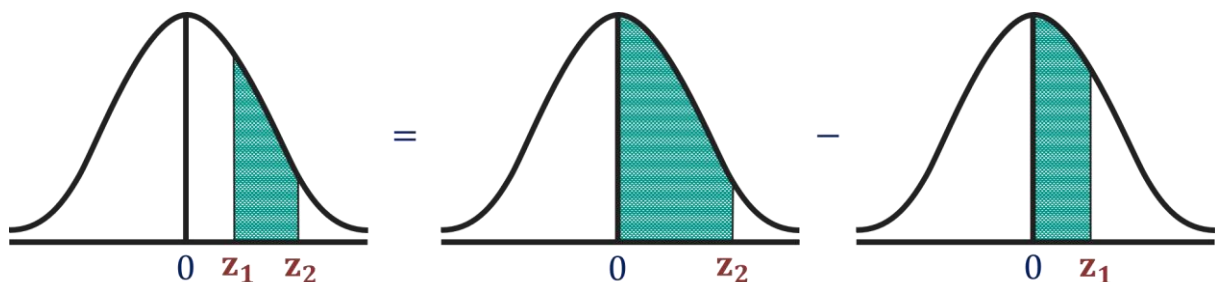
- Area between $Z = z_1$ to $Z = \infty$ which can be represented as $P(Z \geq z_1)$
- Also, $P(Z \geq z_1) = P(Z \leq -z_1)$ as normal distribution is **symmetric** about x - axis.



(4) **Area between $Z = z_1$ to $Z = z_2$, which can be represented as $P(z_1 \leq Z \leq z_2)$.**

→ Now, $P(z_1 \leq Z \leq z_2) = P(0 \leq Z \leq z_2) - P(0 \leq Z \leq z_1)$

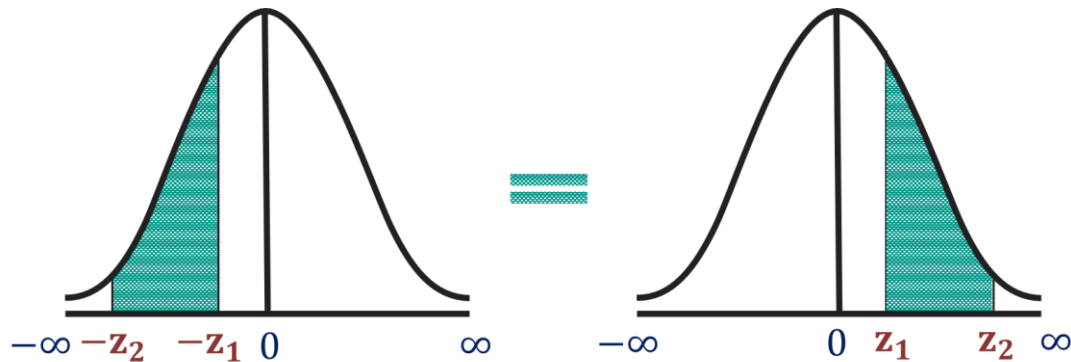
→ Since, $P(0 \leq Z \leq z_1)$ and $P(0 \leq Z \leq z_2)$ can be found by using case – I.



Unit – 3 Probability Distribution

→ **Note:**

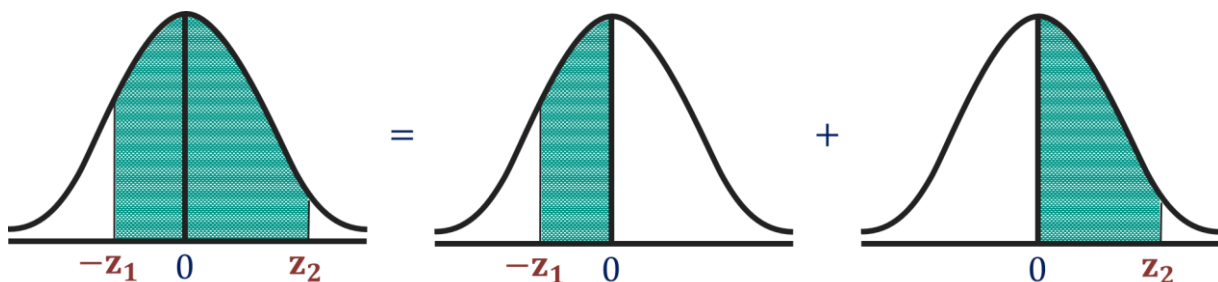
- Area between $Z = -z_2$ to $Z = -z_1$ which can be represented as $P(-z_2 \leq Z \leq -z_1)$
- Also, $(-z_2 \leq Z \leq -z_1) = P(z_1 \leq Z \leq z_2)$ as normal distribution is **symmetric** about x - axis.



(5) Area between $Z = -z_1$ to $Z = z_2$, which can be represented as $P(-z_1 \leq Z \leq z_2)$.

→ Now, $P(-z_1 \leq Z \leq z_2) = P(-z_1 \leq Z \leq 0) + P(0 \leq Z \leq z_2)$

→ Since, $P(-z_1 \leq Z \leq 0)$ and $P(0 \leq Z \leq z_2)$ can be found by using case - I.



Properties of Normal Distribution

- The curve is bell shaped and symmetrical about the line $x = \mu$.
- Mean, median and mode of the distribution coincide. i.e. the normal distribution is symmetrical.

i.e., Mean = Median = Mode

Example of Method-4: Normal Distribution

H	1	If mean is 50 and standard deviation is 10, then find $P(50 \leq X \leq 80)$. $[P(0 < Z < 3) = 0.4987]$ Answer: 0.4987
T	2	If X is a normal variate with a mean of 30 and an SD of 5, find the probability that (i) $26 \leq X \leq 40$, (ii) $X \geq 45$. Given: $P(0 \leq Z \leq 0.8) = 0.2881$, $P(0 \leq Z \leq 2) = 0.4772$ and $P(0 \leq Z \leq 3) = 0.4987$ Answer: (i) 0.7653, (ii) 0.0013
H	3	The breaking strength of cotton fabric is normally distributed with $E(x) = 16$ and $\sigma(x) = 1$. The fabric is said to be good if $x \geq 14$. What is the probability that a fabric chosen at random is good? $[P(0 < Z < 2) = 0.4772]$ Answer: 0.9772
C	4	If X is normally distributed and the mean of X is 12 and the SD is 4. Find out the probability of the following. (a) $X \geq 20$, (b) $X \leq 20$, (c) $0 \leq X \leq 12$ Given: $P(0 < Z < 2) = 0.4772$, $P(0 < Z < 3) = 0.4987$ Answer: (a) 0.0228, (b) 0.9772, (c) 0.4987
H	5	A manufacture knows from his expenditure that the resistance of resistor he produces is normal with $\mu = 100$ ohms and $\sigma = 2$ ohms. What percentage of resistors will have resistance between 98 ohms and 102 ohms? Given: $P(0 < Z < 1) = 0.3413$ Answer: 68.26%
T	6	The average seasonal rainfall in a place is 16 inches with an SD of 4 inches. What is the probability that the rainfall in that place will be between 20 and 24 inches in a year? $[P(0 < Z < 1) = 0.3413$, $P(0 < Z < 2) = 0.4772]$ Answer: 0.1359

Unit – 3 Probability Distribution

H	7	Weights of 500 students of college is normally distributed with $\mu = 95$ lbs. & $\sigma = 7.5$ lbs. Find how many students will have the weight between 100 lbs. and 110 lbs. [$P(0 < Z < 2) = 0.4772$, $P(0 < Z < 0.67) = 0.2486$] Answer: ≈ 114 students
C	8	For a random variable having the normal distribution with $\mu = 18.2$ and $\sigma = 1.25$, find the probabilities that it will take on a value (a) less than 16.5, (b) between 16.5 and 18.8. [$P(0 < Z < 1.36) = 0.4131$, $P(0 < Z < 0.48) = 0.1844$] Answer: (a) 0.0869, (b) 0.5975
C	9	The compressive strength of a sample of cement is normally distributed with mean 6000 kg/cm^2 and standard deviation 100 kg/cm^2. (a) What is the probability that a sample strength is less than 6250 kg/cm^2? (b) What is probability if sample strength is between 5800 and 5900 kg/cm^2? (c) What strength is exceeded by 95% of the samples? [$P(0 < Z < 2.5) = 0.4938$, $P(0 < Z < 1) = 0.3413$, $P(0 < Z < 2) = 0.4772$, $P(0 < Z < 1.65) = 0.4505$] Answer: (a) 0.9938, (b) 0.1359, (c) $\approx 5835 \text{ kg/cm}^2$
C	10	A sample of 100 dry battery cells is tested and found to have an average life of 12 hours and a standard deviation of 3 hours. Assuming the data to be normally distributed, what percentage of battery cells are expected to have life (a) more than 15 hours? (b) less than 6 hours? (c) between 10 and 14 hours? [$P(0 < Z < 1) = 0.3413$, $P(0 < Z < 2) = 0.4772$, $P(0 < Z < 0.67) = 0.2486$] Answer: (a) 15.87%, (b) 2.28%, (c) 49.72%

Unit – 3 Probability Distribution

H	11	The distribution of heights of 1000 soldiers is normal with mean 165 cm and standard deviation 15 cm. Find how many soldiers are of height (a) less than 138 cm? (b) more than 198 cm? (c) between 138 cm and 198 cm? [$P(0 < Z < 1.8) = 0.4641$, $P(0 < Z < 2.2) = 0.4861$] Answer: (a) ≈ 36, (b) ≈ 14, (c) ≈ 950
H	12	In a photographic process, the developing time of prints may be looked upon as a random variable having a normal distribution with mean 16.28 seconds and standard deviation 0.12 seconds. Find the probability that (a) it will take anywhere from 16.00 to 16.50 seconds to develop one of the prints. (b) it will take at least 16.20 seconds to develop one of the prints. (c) it will take at most 16.35 seconds to develop one of the prints. [$P(0 < Z < 1.83) = 0.4664$, $P(0 < Z < 2.33) = 0.4901$] [$P(0 < Z < 0.67) = 0.2486$, $P(0 < Z < 0.58) = 0.2190$] Answer: (a) 0.9565, (b) 0.7486, (c) 0.7190
H	13	The customer accounts of a certain department store have an average balance of ₹120 and a standard deviation of ₹40. Assume that account balances are normally distributed. Find the proportion of accounts (a) over ₹150. (b) between ₹100 and ₹150. (c) between ₹60 and ₹90. [$P(0 < Z < 0.75) = 0.2734$, $P(0 < Z < 0.5) = 0.1915$, $P(0 < Z < 1.5) = 0.4332$] Answer: (a) 0.2266, (b) 0.4649, (c) 0.1598

Unit – 3 Probability Distribution

C	14	Assuming that the diameters of 1000 brass plugs taken consecutively from a machine form a normal distribution with mean 0.7515 cm and standard deviation 0.002 cm. Find the number of plugs likely to be rejected if the approved diameter is 0.752 ± 0.004 cm. [$P(0 < Z < 1.75) = 0.4599$, $P(0 < Z < 2.25) = 0.4878$] Answer: ≈ 52
H	15	Two machines A and B produce items whose weights are normally distributed. Machine A : $\mu = 500$ g, $\sigma = 25$ g, Machine B : $\mu = 500$ g, $\sigma = 40$ g Out of 600 items from each machine, find the expected number of items weighing more than 540 g. [$P(0 < Z < 1.6) = 0.4452$, $P(0 < Z < 1) = 0.3413$] Answer: From Machine A ≈ 33, From Machine B ≈ 95
T	16	If Z is a standard normal variable and z_1 is Z-score, then obtain the values of z_1 satisfying the following conditions: a) $P(-1 \leq Z \leq z_1) = 0.5255$ b) $P(z_2 \leq Z \leq 2) = 0.7585$ [$P(0 < Z < 1) = 0.3413$, $P(0 < Z < 2) = 0.4772$] Answer: $z_1 \approx 0.48$, $z_2 \approx -0.78$
C	17	In an examination, minimum 40 marks are required for passing and 75 marks for distinction. If 45% of the students passed and 9% obtained distinction, find the mean and standard deviation of the distribution of marks. [$P(0 < Z < 0.13) = 0.0517$, $P(0 < Z < 1.34) = 0.4099$] Answer: $\mu = 36.24$ Marks, $\sigma = 28.93$ Marks
H	18	Let X be a normal random variable. Given that $\sigma = 10$ and $P(X \leq 35) = 0.0668$. Find the mean μ. [$P(-1.5 < Z < 0) = 0.4332$] Answer: $\mu = 50$

Unit – 3 Probability Distribution

T	19	The marks obtained out of 70 by students in a test are normally distributed with mean 55 and standard deviation 10. Find the probability that a student does not score between 52 and 68 marks. Given: $P(0 < Z < 0.3) = 0.1179$, $P(-1.3 < Z < 0) = 0.4032$. Answer: 0.4789
T	20	The probability density function of a normal random variable is $f(x) = \frac{1}{4\sqrt{2\pi}} e^{-\frac{1}{32}(x-50)^2}; -\infty < x < \infty$ Obtain parameters of this distribution and find the values of following: a) $P(52 \leq X \leq 58)$ and $P(X - 45 \leq 4)$ Given: $P(0 < Z < 2) = 0.4772$, $P(0 < Z < 0.25) = 0.0987$ $P(0 < Z < 2.25) = 0.4878$ Answer: $\mu = 50$, $\sigma = 4$, (a) 0.2857, (b) 0.3891
T	21	A unicorn AI startup receives applications from 120,000 software engineers worldwide. Each applicant completes a machine-learning aptitude assessment. Historical data show that the final scores are approximately normally distributed with mean 68 and standard deviation 11. - The company interviews only the top 1.5% of applicants. - Applicants scoring below 45 are automatically rejected. Find: (a) The minimum score required to receive an interview invitation. (b) Approximately how many applicants will be automatically rejected? (c) The probability that a randomly selected applicant scores high enough for an interview, given that they were not automatically rejected. Given: $P(0 < Z < 2.17) = 0.4850$, $P(0 < Z < 2.09) = 0.4817$ Answer: (a) ≈ 92, (b) 2196, (c) 0.0153

Unit – 3 Probability Distribution

T	22	A space agency is preparing a mission to Mars. Every astronaut carries an emergency oxygen cylinder. The operating duration of the cylinders is approximately normally distributed with mean 11.8 hours and standard deviation 0.6 hour. Mission engineers have established the following safety rules: • Cylinders lasting less than 10.5 hours are unsafe. • Cylinders lasting more than 13 hours are premium quality. • A production batch contains 18,000 cylinders. • Each astronaut carries two cylinders chosen independently. Find: (a) Estimate the number of unsafe cylinders in the batch. (b) What is the probability that both cylinders assigned to an astronaut are premium quality? (c) What is the probability that at least one of the two cylinders is unsafe? Given: $P(0 < Z < 2.17) = 0.4850$, $P(0 < Z < 2) = 0.4772$ Answer: (a) 270, (b) ≈ 0.00052, (c) ≈ 0.0298
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***** End of the Unit *****